GlucoTwin India: architecture

Static record + live wearable stream -> personalized twin -> spike forecast and what-if

1 Data (synthetic only)

Static stream: EHR

age, sex, BMI, duration
HbA1c, eGFR, LDL, TG
TCF7L2 genotype, adherence
diet pattern, region, activity

Dynamic stream: wearables

CGM glucose (5 min)
heart rate, steps, sleep
Indian meal log: carbs, GI, glycemic load
sensor gaps, unlogged meals

Coupled through insulin resistance: EHR traits scale every meal response

2 Fusion and models

Feature builder

meal-time snapshot (no leakage):
CGM level + trend, HR, steps,
last night's sleep, previous meal,
static EHR features

A Spike model (LightGBM)

P(rise >= 60 mg/dL within 2 h)
+ expected rise (regressor)
patient-level train/val/test split
baselines + fusion ablation

B Digital twin

state: patient response gain
HbA1c prior -> Kalman update
after every observed meal
95% credible band

What-if simulator

meal swap (rice -> millet)
portion size, post-meal walk,
sleep hours
forecast curve + uncertainty band

3 Service and UI

FastAPI

/patients /advance
/whatif /model-card
streaming cursor

Doctor dashboard

virtual patient view
plan A vs plan B
twin state + recent meals
evidence panel

Loop: each logged meal is observed 2 h later -> twin gain updates -> next forecast is sharper

Privacy: no real patient data is used. Docker build generates the cohort (seed 42), trains and evaluates; the container serves API and dashboard on :8000